ISHAN P. BORKER

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Profile Summary

- Engineering Lead , Machine Learning Competency
- 9.16 years of total experience which include 8.16 years of ML, GenAI experience
- Purposeful software development professional with over 9.16 years of comprehensive experience across cutting-edge platforms, including Generative AI, Data Science, Machine Learning, NLP, Computer Vision, and Network Service Orchestration.
- Demonstrated expertise in requirement analysis, solution architecture, and the delivery of robust, scalable, end-to-end applications. Proficient in a diverse technology stack, including Python, C++, Golang, Docker, Linux/Unix, POSTMAN, and JSON scripting.
- With over 9.16 years of specialized focus in Generative AI, Computer Vision, and advanced machine learning, I have successfully led projects from pre-sales and proof-of-concept stages through to deployment and ongoing support.
- My background includes research and development, networking, and the delivery of customized solutions that address unique client needs.
- Highly adaptable and collaborative, I excel in solving complex business challenges, embracing emerging technologies, and performing in fast-paced, dynamic environments.
- I am committed to continuous learning and am always ready to take on challenging roles that drive innovation and deliver measurable business value.

Work Experience

Persistent Systems Limited, Goa, India – May 2022 – Present, Engineering Lead

Role: Jan 2026 – Present as Engineering Lead

Project: UHG Optum Insights – Data: Dojo360 AI

Client: United Health Group

Role: Generative AI Engineer

Responsibilities:

[DX] E2E Testing of MCP Server post migration to the new Agentic Gateway: Executed end-to-end testing of Dojo360 MCP server following migration to the new Agentic Gateway, validating key MCP endpoints through Insomnia including listTools, listResources, listPrompts, listResourceTemplates, listmodules, searchprofile, searchmodule. Established secure connectivity between Dojo360 MCP Client and Agentic Gateway Production environment, generated and verified authentication tokens and successfully validated the list of Tools like ListDojo360 Modules, Search Dojo360 Documentation, Get Dojo360 Module Details, Dojo360 Module Profile Details, Dojo360 Module Changelog and Dojo360 Module Upgrade Guide through Insomnia

[CMA] Setup Alarms in Splunk for Code Agent: Worked on Splunk On-Call alerting, workflow failure monitoring, severity-based notifications and configurable alert policies across CodeAgent. Added comprehensive unit tests for key failure scenarios including API errors, unauthorized access, service outages, Terraform timeouts and validation failures. Completed end-to-end validation using Insomnia and Splunk-On Call Dashboard

DocSearch: Upgrade LLM to GPT-5: Upgraded DocSearch LLM deployment to GPT-5 and GPT-5.4-mini including configuration, code and deployment updates. Successfully validated /ai/search and /ai/generate APIs, completed testing in Insomnia and in dev environment in GitHub Actions and merged PR to the main branch

Add CodeAgent API to https://api-dojo360.optum.com: Implemented CodeAgent API integration using FastAPI including API routing, configuration management, JWT Authentication, token validation and OpenAPI enhancements; Enabled and validated Swagger UI and OpenAPI endpoints for codeagent docs through Azure Function Apps; Developed authentication components like jwt auth, verify access token, added unit tests for JWT fallback and JWKS validation and ensured successful test execution; Configured Azure Front Door

Key Skills and Knowledge

Generative AI Technologies

RAG
LangChain
LLM
AWS Chatbot
Amazon Bedrock
BrowserBase
Perplexity.AI
Cline
Stagehand
Amazon Kendra
Amazon OpenSearch
Amazon Q Business
Amazon Q Plugin
Mesop
GenRocket
MCP
FastMCP
Agents
AgentSkills
YAML
Azure Functions
Terraform

Machine Learning Technologies

Python
FastAPI
Data Science
Machine Learning
Computer Vision – OpenCV
NLP – FastText
spaCy
Sci-kit learn
NumPy
Pandas
Linear and Logistic Regression
Clustering algorithms
Decision Trees
Neural Networks
Reinforcement Learning
Feature engineering
statistical analysis

routing, URL reqrites and Terraform updates in dev environment; Updated OpenAPI security schemas, documentation handlers, environment configurations, dependencies and deployment artifacts; Extended similar API, Swagger, Authentication and Front Door integrations to dojo360-ai-docs-search service and updated front door terraform code in platform-infrastructure route mappings; Addressed PR feedback, supported EPL-CI workflow execution, deployed front door changes to dev environment

[DX] Implement and publish new customer-facing Dojo360 skills: Developed and published customer-facing Dojo360 onboarding skills by creating skill manifests, prompts, instructions, smoke tests enabling streamlined user onboarding; Built deployment automation with deployment/rollback scripts, pipeline creation runbooks; Enhanced documentation, operational guidance and local execution support through comprehensive README updates and runbook creation; Designed and implemented HCPCTL-based pipeline bootstrap solution for automating ephemeral account onboarding and pipeline creation workflows; Collaborated with stakeholders to address PR feedback, validate solutions and drive publishing readiness

[NxG] Validate NextGen compatibility in AWS ephemeral accounts: Validated NextGen compatibility in AWS ephemeral account by provisioning and verifying VPCs, Subnets and Security Groups using Terraform, AWS Portal and HCP Console; Automated AWS infrastructure deployment through Terraform and GitHub Actions, updating and executing terraform-plan.yml and infrastructure.yml workflows; Enhanced Terraform configurations for AWS networking and bootstrap automation; Implemented secure credential management using vaults, secrets and bootstrap modules supporting automated infrastructure provisioning; raised PR, validated acceptance criteria, delivered demo to stakeholders, drove process-improvement discussions and published comprehensive technical documentation

[NxG] Validate NextGen compatibility in Azure ephemeral accounts: Validated NextGen compatibility in Azure ephemeral account by provisioning and verifying VNets and Subnets using Terraform and Azure Portal, automated Azure infrastructure deployment through Terraform and GitHub Actions by updating and executing terraform-plan.yml and infrastructure.yml workflows; Enhanced infrastructure modules and Terraform configurations for Azure networking and subscription bootstrap; Implemented secure cloud deployment practices using Azure Key Vault, secret management and GitHub repository integrations; raised PRs, validated acceptance criteria, resolved deployment issues and published technical documentation

[DX] Compare MCP with Skills and provide recommendations: Designed and implemented a Terraform evaluation framework to compare MCP and Agent Skills approaches by enhancing agent metadata, tools and execution workflows; Established a structured evaluation pipeline (init -> plan -> apply -> destroy) with artifactory capture (logs, state, plan outputs) and standardized failure tracking ; Developed reusable prompts, instructions and skill templates with consistent YAML frontmatter to streamline agent, instruction and prompt creation; Covered automated evidence collection, failure documentation, comparison and reporting of infrastructure-as-code execution quality; Completed communication of skills created using MCP server and prepared detailed documentation covering points: comparison of Terraform Code Generation using MCP vs AgentSkills, guidance on when to use MCP vs AgentSkills

[Skills] Develop Mechanism to update skills seamlessly: Designed and implemented a modular Agent Skills Framework aligned with the Agent Skills open standard to enable seamless skill updates without service restarts; built a dynamic skills loader for auto-discovery and execution, implemented MCP + Terraform setup, evaluated developer experience (auto-completion, error handling, performance), and documented framework, validation pipelines, versioning controls, CI/CD integration, centralized skills catalog (skills_catalog.yml) with metadata/governance, fallback handling, VS Code onboarding, testing suites, role-based access control, centralized hosting, monitoring, and performance tracking to ensure scalable, maintainable skill lifecycle management; Developed a standardized Skills Management Framework and centralized skills catalog (skills_catalog.yml) with structured metadata, governance, dynamic loading, filtering (tags, team, status) and fallback handling; implemented unified execution, validation pipelines, versioning controls, CI/CD quality checks and registry sync, improved developer experience via onboarding guides, VS Code integration and testing suites, enforced role-based access control, and established centralized hosting/monitoring for DX logs and performance to deliver enterprise-ready skill lifecycle management without agent restarts; Implemented the mechanism for automated skill updates and enhanced automation within the MCP server and prepared detailed documentation covering points: steps to set up Skill for MCP, reusability of skills to improve developer productivity, comparative summary and insights between MCP and Skills

[TF Eval] [SPIKE] Validation Test Suite for Generated Terraform Code and Verify Full Stack Generation through TF Validation: Conducted research to evaluate a validation test suite for generated Terraform code, analyzing the Terraform Agent Test framework, proposing implementation strategies and authoring design documentation with technical recommendations; Implemented a comprehensive validation test suite for generated Terraform code using the Terraform Agent Test framework; designed and executed a multi-cloud

Machine Learning Libraries

Tensorflow
Keras
SpaCy
BERTopic
NLTK
PyPDF
PyLDAvis
PySpark
MLFlow

Algorithms

Classification
Regression
Clustering
Time series forecasting
Topic Modelling
Latent Dirichlet Allocation
Yolov3
Yolov4
Mask R-CNN
Faster R-CNN
OpenCV
Fast text
Named Entity Recognition (NER)
Relationship Extraction (REL)

Other Skills

Actionable Web Technologies
JSON
HTML
JS
PHP
Linux/Unix Commands
Curl Commands
SQL
MySQL
Amazon S3
SpagoBI
Tableau
Navicat
GNU-Plot
C++
Golang
Amazon Aurora RDS
MongoDB Compass

OS

Linux/Unix
Windows
macOS

Python Libraries

(Azure, AWS, GCP) test bed covering multi-stack provisioning (App Service, Redis/Mem Cache, Cloudflare CDN), automated GitHub Actions workflows (terraform-plan, bootstrap), and end-to-end testing via ephemeral repositories and isolated branches—generated code through Insomnia, deployed across CSPs, validated terraform plan/apply for services (AWS ECS, Azure App Service/Function App), ensured correct tfstate and bootstrap execution, and tracked results in structured scoring sheets

[MCP] Add workflows for MCP registry update: Implemented MCP registry publishing workflow by adding mcp-spec.json and publish-to-mcp-registry.yml to GitHub repository; updated ownership metadata and subdomain, successfully merged PR and executed the workflow to publish the MCP server

[DX] Add tools and skill for Modules upgrade guide and changelog: Developed and integrated an MCP tool to generate module upgrade guides and changelogs using MongoDB-backed search, content parsing and JSON-RPC responses; implemented robust error handling and user prompts aligned with the standardized MCP dispatch pipeline; Refactored to use the shared Database Client (no direct MongoClient), made configurations fully config-driven and increased the unit test coverage to 80% based on PR reviews

[DX] Add Dojo360 MCP to HCP API Gateway: Added AI MCP endpoint in platform API Gateway (YAML); updated front door in dojo360-platform-infrastructure Terraform (frontdoor local/prod/dev). Merged to main branch and deployed via GitHub. Collaborated with Agentic Gateway team to update MCP server domain; added recommended tags to dojo360-ai-mcpserver-func-cus-dev resources in dojo360-platform-infrastructure; redeployed to dev, QA, and prod environments

CodeAgent & MCP Server Interaction: Created CodeAgent endpoint in dojo360-ai-code-agent repository with MCP client wrapper; tested payloads in Insomnia. Added unit tests (all passed) and end-to-end testing with negative/edge cases; presented client demo. Added MCP wrapper thin layer, error handling, logging, and retry logic to dojo360-ai-code-agent; created .md prompts for agentic profile selection (service names, Insomnia bodies, regex search from MongoDB without examples). Updated searcher/agent code to validate search API; completed negative/edge Insomnia tests

[Tool: Modules Docs] Develop tool to retrieve profile documentation and examples: Built tool to retrieve profile docs/examples: MCP docs tool queries CSP-specific modules_catalog in MongoDB for metadata. Added/tested API endpoints getProfileDocs/getProfileExamples in Insomnia for all modules/profiles per CSP. Identified correct profiles for validation; MCP client tested endpoints; MCP handled DB interactions via function app/tool decisioning. Tested filtering by module/profile from href; delivered client demo. Refactored code based on dojo360-ai-mcp-server changes using FastMCP; tested all API endpoints, added unit tests, improved coverage to 80%

MCP Client & Server Integration Testing: Conducted comprehensive integration testing for all MCP client API functions using Insomnia, covering unit tests, negative/edge cases. Verified end-to-end API endpoints; explored dojo360-ai-mcp-client-tool repository and MCP connection commands. Configured MongoDB connection; ingested docs/module catalog data into dojo360_docs database via Azure Blob Storage (JSON to embeddings)

Infrastructure & Documentation, Whitelist NextGen IP range for MCP Server: Collaborated with Enterprise Tech Infra to whitelist static/queue sites and NextGen MCP server IPs (bypass Azure), prepared and uploaded documentation to Dojo360 Wiki, activated Azure PIM, updated Function App ingress and integrated Azure Hub Commons Terraform; updated dynamic tags in tfvars across dojo360-ai repos, refreshed CI/CD workflows (CI.yml, CD.yml, create-release.yml, release-promotion.yml, copilot-setup-steps.yml) to ensure passing build checks and merged PRs to main, tested CMA JSON requests for AWS/Azure in Insomnia to validate profile picking and CSP mapping, fixed bootstrap and tfstate management workflows, reviewed generated Terraform code for accuracy, and resolved Dependabot/CodeQL vulnerabilities by updating dependencies and refining workflows

[Tool: Modules Catalog] Generate CSP Specific list of Modules, Profiles & Develop Modules Catalog Tool to Dojo360 MCP Server: Ingested modules catalog docs into MongoDB; performed local Insomnia testing (negative/edge cases) and raised PR. Built Browse Catalog tool for dojo360-ai-mcp-server: lists modules by technology (container/compute/database), exposes profiles, supports decisioning; backed by local MongoDB mock data. Implemented in VSCode Jupyter Notebook and server repository; reworked for CSPs (3 separate tools: one per CSP) using modular code (without Jupyter); updated CSV in modules_catalog collection. Added RAG pipeline unit tests; verified via Amazon Q LLM prompts ("list modules/profiles"); delivered client demo

Agentic MCP & Code Generator Enhancements: Developed Agentic MCP program accepting math expressions, using UAIS LLM for tool decisioning (add/subtract); built in VSCode Jupyter Notebook. Enhanced Code Generator Agent via Agentic Gateway: removed Tool Poisoning attack; tested APIs in Insomnia. Built MCP tool invoking LLM via FastMCP/Agents; compared responses in VSCode Jupyter Notebook

Dojo360-ai-feedback-api:Added and deployed the healthcheck endpoint, successfully validated via GitHub Actions and Insomnia; Deployed feedback api functionapp with managed identity for storage account;

Numpy
Pandas
SciPy
Seaborn
Scikit-learn
Plotly-dash
Matplotlib

Tools/Environments

EMS
Crosswork Workflow Manager (CWM)
Network Service Orchestration (NSO)
Lux
SonarQube
Jupyter Notebook
Google Colaboratory
Kaggle notebook
Databricks Notebook
Amazon SageMaker Notebook
Spyder
PyCharm
VSCode
POSTMAN
Swagger UI
GitHub
AWS EC2
Azure ML Studio
GitHub Copilot
Microsoft Copilot
AWS Code Whisperer
Amazon Q LLM
Insomnia
Microsoft Azure
CI/CD
GitHub Actions

Hobbies

- Playing indoor and outdoor games
- Gymming, Yoga
- Listening to music
- Reading books, newspapers and technical magazines
- Watching news and movies
- Watering plants

Resolved security vulnerabilities for dojo360-ai-docs-ingestion and dojo360-ai-feedback-api, pushed the code changes in qa environment (Promotion Run CI/CD) and performed the validation

Tools/Technologies: MCP, Insomnia, MongoDB Compass, GitHub Copilot, Microsoft Copilot, Agents, FastMCP, VS Code Jupyter Notebook, Amazon Q LLM, Microsoft Azure, Azure Functions, Terraform, AgentSkills, YAML, CI/CD, GitHub Actions

Role: Apr 2024 – Dec 2025 as Lead Software Engineer

Project: UHG Optum Insights - Data (Onsite) – IBM CMOD

Client: United Health Group
Role: Generative AI Engineer
Responsibilities:
Created evaluation templates for LLMs in Amazon Bedrock and documented AWS search service requirements, implemented lexical and semantic search in OpenSearch; handled claims data search and ingestion using Kendra and Q Business with S3 integration, compared search results from Kendra, Q Business, and OpenSearch; documented findings in tables and reports, explored Amazon Q plugin architecture in OpenSearch UI; set up Q-business index as a data source and provided accuracy feedback, configured Q-business index, demonstrated search scenarios, created custom plugins, and linked OpenSearch indices to S3 datasets; converted datasets to OpenAPI formats, assessed and compared search performance and latency across OpenSearch, Kendra, and Q Business using complex queries, prepared customer demo presentations, worked on image ingestion into OpenSearch by connecting S3 bucket of X-ray dataset to LLM, converting DICOM dataset into jpeg for ingestion, tested knowledge base of this dataset to Amazon Bedrock, developed front end UI for Aurora RDS database for patient records using mesop (insert and search operation), migrated the same code to Amazon SageMaker Notebook
SOW1.0: Worked on Gen AI Tool and Milestone Mapping, explored the migration approach from DB2 to AuroraDB, and activities utilizing AWS Transform for acceleration, automation, and standardization, designed end-to-end architecture diagram for Query Pathway, evaluated and compared GenRocket versus custom Python approaches for large-scale, heterogeneous test data generation, documented requirements, explored generating and loading bulk data in GenRocket directly to Aurora, explored on the set up of GenRocket for storing EDI files in AWS S3 from an EC2 instance, configured GenRocket for bulk data generation and XML EDI output, and stored generated EDI data (single/bulk) in S3 from local system for business scenarios, exploring simultaneous data generation in S3 and uploading to AuroraDB using GenRocket

Tools/Technologies: Amazon Kendra, Amazon Q Business, Amazon OpenSearch, OpenSearch Dashboard, S3, Amazon Q Plugin, Amazon SageMaker Notebook, Amazon Aurora RDS, mesop, GenRocket

Project: Generative AI Platform development - Co-Engineering – Data

Client: Baldwin Risk Partners
Role: Generative AI Engineer
Responsibilities:
Classification: Executed RAG classification and extraction code for different policy documents. Explored chunking strategy for large tables obtained after extracting documents using LLM calls
Integration of Amazon Bedrock Agent and AWS Chatbot with Microsoft Teams: Changed email while testing a message in Amazon SNS and publishing it inside Teams Channel, created a knowledge base and Amazon Bedrock Agents and tested it with prompts, created an application inside developer portal in Microsoft Teams for integration
Actionable Web Technologies: Created a tabular analysis for evaluating different actionable web technology tools like BrowserBase, Maxun based on future scalability, and browser dependencies. Python code to enter URL and downloaded relevant PDFs and using BrowserBase packages like Playwright and its APIs to send URL to download relevant PDFs. Worked on stealth mode by modifying existing codebase to bypass captcha or login methods and capture metadata. Executed the same code on different carriers being implemented under Carrier Document Download, downloading policy document for each agency code using the

credentials for CNA, Selective Flood carrier websites, and integrating metadata to it, worked on downloading policy document using LLM prompts and Stagehand approach on Hartford carrier website
Capabilities of Perplexity.AI: Improved prompt caching, dynamic updates, response accuracy, cache persistence and query handling using perplexity.ai, retrieving response format obtained from perplexity API. Tested feasibility of Perplexity in Copilot UAT and potential business benefits
Mosaic AI Model Fine-Tuning: Performed model fine-tuning on RAG Component Classification and Extraction dataset from S3 bucket using Databricks Notebook, model testing using gpt-35-turbo-SLM, trained fine-tuning run using meta-llama models, deployed fine-tuned model to serve endpoint and testing it
Worked on client requirements like risk map automations, combination of structured and unstructured sources, graph DB for automating RFP process, LLMs with structured knowledge graphs, graph RAG, knowledge augmented graph and prepared documentation. Prepared use case presentations for all POCs
Tools/Technologies: RAG, LangChain, LLM, AWS Chatbot, Amazon Bedrock, BrowserBase, Perplexity.AI, Actionable Web Technologies, Databricks Notebook, Mosaic AI, pyspark, mlflow, Playwright, Stagehand

Project: CISCO NSO Engineering

Client: CISCO
Role: Software Engineer
Responsibilities:
Worked on EMS adapter activity support for different workflows like distribute, activate, commit image on IOS-XE device, working on API hardening: adding error handling conditions to the EMS adapter Go code, validating the inputs for SWIM APIs, performed code refactoring on it, performed add image, activate and commit using install method for IOS-XE device, updating the EMS adapter to be compliant with CW infrastructure by refactoring the code using Authentication, No Authentication, running different EMS APIs inside cw-admin SVM Infrastructure like import, activate, distribute and commit image using curl commands and updating the wiki page, worked on the unit test cases for the refactored codes and generated test coverage report, developed POC for Go client for fetching the CCO catalog and SMU details for each record in catalog, fetched multiple devices inside the jobsummary EMS Code
Tools/Technologies: EMS, SWIM, CWM, Go

Role: May 2022 – Mar 2024 as Senior Software Engineer

Project: Routermation CISCO NSO Offshore Transition

Client: CISCO
Role: Software Engineer
Responsibilities:
Worked on Crosswork Workflow Manager (CWM) installation, creation of adapters, resources, secrets, EPNM, created workflows, adapters for different applications and tested it on POSTMAN, Captured requests and responses for different APIs like import, distribute, activate and commit image, get job summary, get device details, device upgrade, consolidated workflow etc. for Element Management Systems (EMS) on POSTMAN, implemented codes for networking device upgrade and created adapters using Golang and executed the workflows in CWM UI, executing the SWIM APIs in CWM UI instance, worked on Network Service Orchestration (NSO) installation and Routermation Automation Kit for device onboarding, fleet upgrade and golden config, performed LNT device upgrade using RAKfu Main workflow and lux test cases, fixed critical and major issues for CW-EMS adapter using SonarQube, updated the CW-EMS adapter with CWM 1.1 support and library integration, worked on IOS-XE device upgrade
Tools/Technologies: EMS, SWIM, CWM, Go, POSTMAN, NSO, Lux

Project: PDF Text Extraction

Client: IBM Infosphere
Role: AI Engineer
Responsibilities:
Worked on PDFs from URLs. Extracted, and preprocessed the text from it using PyPDF and generated histograms using nltk. Worked on extractive text summarization using different summarizers like LSA, Luhn, Text Rank, and Lex Rank and calculated the inference time of each summarizer. Tested FastAPI codes for training and inference concerning Fasttext classifier on text classification dataset with categories like Chemical Hazardous, Restricted Prohibited, Neither, calculated the total training time of all the models. Performed Volumetric Stats for Catalog files like unique UNSPSC for Primary, Corrected Category, number of data points available and missing for each UNSPSC for corrected category

Tools/Technologies: PyCharm, Spyder, FastAPI, PyPDF, NLTK, Text Summarizers

Project: Serving NER and REL models using SpaCy

Client: Chia Healthcare
Role: AI Engineer
Responsibilities:
Worked on clinical trials dataset analysis and calculating performance metrics scores using SpaCy for Named Entity Recognition (NER) and Relationship Extraction (REL) using pipelines like tok2vec (on CPU), different transformers (on GPU), plotted loss curves for NER, REL, combined models. Improving performance metrics like precision, and f1 score for all models. Served the English SpaCy model on FASTAPI and tested it on POSTMAN. Served own trained NER, and REL models using FASTAPI code and tested the model on POSTMAN and in Swagger UI. Deployed model on AWS EC2 T2 medium instance, dockerized FASTAPI NER code, and pushed the docker container on GitHub
Tools/Technologies: Python, SpaCy, NER, REL, Google Colaboratory, FASTAPI, AWS EC2, GitHub, POSTMAN

Project: Topic Modelling SEC filings

Client: FinTech Companies
Role: AI Engineer
Responsibilities:
Worked on topic extraction on docoh dataset of companies (Intuit, UHG, Wells Fargo, etc.) using packages like BERTopic, Latent Dirichlet Allocation (LDA), Contextual Topic Modelling (CTM). Performed dimensionality reduction (PCA, UMAP), and clustering (HDBSCAN) techniques, using transformers like Roberta, finbert, and sentence transformers to improve topic extraction and visualizing the topics using PyLDAvis. Development is done using Python inside Google Colaboratory
Tools/Technologies: LDA, BERTopic, CTM, PCA, UMAP, PyLDAVis, Google Colaboratory, Python

Other Projects

Binary classification of network attacks:
Worked on binary classification of network attacks present in NSL-KDD dataset, performed EDA, data pre-processing, generating plots to understand different features using matplotlib, seaborn, found statistical insights, performed feature engineering using one-hot encoding, label encoding, feature scaling, normalization, performed feature selection using methods like filter, wrapper and embedded, performed model selection using cross-validation, bootstrapping, used different machine learning classifiers like Random Forest, Decision Tree, XGBoost, Gradient Boosting, Voting Classifier and trained the model, performed hyper-parameter tuning using GridSearchCV and RandomizedSearchCV, evaluated performance metrics, saved the trained model and exported it on the server using joblib, worked on model explainability using eli5, Partial Dependence Plot, Individual Conditional Expectations, Local Interpretation of Model Explanation, SHAP

Smart Interviewer: Developed MCQ part of the test using Python, created a dataset of question bank using .toml for AI Technothon event. Development is done using Python, Github Copilot, and VS Code

Image Segmentation: Worked on Medical Image segmentation using U-NET, Mask R-CNN on NVIDIA GPU Cloud (NGC)

TechnoPro India, Bengaluru, India - Dec 2020 – Apr 2022, Machine Learning Engineer

- **Annotation (Client: Nexar, Role: AI Engineer)**: Worked on semi-automated annotation using CVAT and ML algorithms like YOLOv3, Faster R-CNN, Mask R-CNN, and manual annotation using LabelBox, LabelImg, GTStudio, led the team of 10 annotators, and provided training on annotations like bounding box, polygon, polyline, landmark, image segmentation, cuboid, video, 3D annotations, instance, and semantic segmentation
- **Smart Farming (Role: AI Engineer)**: Implemented tomato image classification based on image data using SVM, KNN, MLP, and Logistic Regression. Development is done using Python inside the Jupyter lab
- **Reviewer Analysis (Client: Nexar, Role: AI Engineer)**: Analyzed reviewer's incident data using Python in graphs and video dashboards inside Jupyter Notebook using matplotlib, seaborn
- **Industry 4.0 (Client: Toyota Kirloskar Motors, Role: AI Engineer):** Worked on detecting car parts in images and training models using YOLOv3, performed image annotation using VoTT and car parts damage

detection using detectron2, Mask R-CNN models. Worked on predictive bearing maintenance to find the remaining helpful lifetime using regressors like SVR, RF, and DT Regressor. Worked on Object Detection and Tracking of real-time intrusion using YOLOv4, and OpenCV. Development is done using Python inside Google Colaboratory

- **Perimeter Intrusion Detection System (Client: Bengaluru International Airport Limited, Role: AI Engineer)**: Worked on POC, designing RFP on perimeter intrusion detection project for Airport surveillance covering aspects of Machine learning, Computer vision, RADAR for video surveillance, verification, tracking, access control security, and Smart Command and control system, designed presentations, case studies, RFP, POC for different AI-ML projects

Molecular Connections, Bengaluru, India - Jul 2020 – Oct 2020, Senior Software Engineer

- **EHR Predictive Modelling (Role: AI Engineer)**: Cleaned EHR data, performed EDA, visualization, and correlation, performed binary encoding on the lab values, trained the model to calculate performance metrics, exported model on the server using joblib/pickle, explored plotly-dash. Development is done using Python inside Google Colaboratory
- **FHIR HL 7(Role: AI Engineer)**: Implemented Formal Concept Analysis, JSON files parsing, FHIR Parser using Python, worked on HAPI FHIR Server by creating resources from JSON, tested REST operations on POSTMAN, created own FHIR Server

Rubixe, Bengaluru, India - Dec 2019 – Jun 2020, Data Science Consultant

- **Employee Performance Analysis (Role: AI Engineer)**: Cleaned the data, performed EDA, created a visualization to find department-wise performance of employees, and built a model to improve performance using machine learning algorithms like Random Forest, Gradient Boosting, XGBoost, ANN, KNN, Logistic Regression, SVM, Decision Tree with the highest accuracy of 96% using Gradient Boosting, used feature engineering, Grid & Randomized Search CV, SMOTE
- **Spare Parts Inventory Management (Role: AI Engineer)**: Cleaned, analyzed the data, created the visualization, built a model using time series algorithms, achieved better predictions for ARIMA
- **Improving ITSM (Role: AI Engineer)**: Imported data from the server, performed EDA, implemented prediction on priority tickets, reassigned tickets using ML algorithms with an accuracy of 86%, implemented ARIMA, and SARIMA on incidents, achieved better predictions using Rolling Forecast
- **Sales Prediction (Role: AI Engineer)**: Used SMOTE to handle imbalanced datasets, PCA for dimensionality reduction, predicted status of sales potential using classification algorithms, achieved 72% accuracy, plotted ROC, precision-recall curve
- **Telecom Churn Prediction (Role: AI Engineer)**: Predicted churn using ML classification algorithms, achieved an accuracy of 97%, used K-fold, Stratified K-Folds CV to improve performance, plotted ROC, precision-recall curve, calculated churn-risk score, exported model on the server
- **Bank Good Credit Score (Role: AI Engineer)**: Imported Customer Accounts, Demographics, and Enquiry data from the server, created a visualization to check factors influencing customers having a good or bad credit history, and achieved 95% accuracy using ML classification algorithms. Development is done using Python inside Jupyter Notebook for all these projects

Digiapt Software, Bengaluru, India - Oct 2019 - Nov 2019, Associate Software Developer

- Used Tableau to categorize companies' data from Tofler, Zaubacorp, and MCA, and plotted their location using My Maps, used Snovio to capture emails of employees and analyze roles

Eaglys Inc. (Deputed from Teczuno), Tokyo, Japan - Dec 2018 – Jul 2019, Research Engineer

- **Back Office System for Manhour Management (Role: Software Engineer)**: Captured attendance spreadsheet from Google Drive using Python, generated reports in the form of graphs, presentation
- **SecureDB (Role: Software Engineer)**: Tested SQL queries, achieved API documentation using Natural Docs, PyDoc, worked keyword searching on the encrypted database using Python, C++, worked homomorphic encryption for matrix multiplication, addition of numbers using HElib, C++ in docker

Teczuno Global India, Bengaluru, India - Apr 2018 – Aug 2018, Data Analyst

- Gathered data from the government website, helped in application development to locate startup companies using Tableau, and built-up concepts as a part of pre-sales support

Anant Infomedia, Goa, India - Aug 2017 – Feb 2018, Project Trainee

- **Predicting employee attrition using Azure ML Studio (Role: Project Engineer)**: Designed models using classification, regression, clustering, anomaly detection
- **Business Intelligence (Role: Project Engineer)**: Used MySQL, and SpagoBI for Data Analysis, created dashboards, cockpits, and OLAP and deployed on SpagoBI server, worked on PHP, JS, HTML, Navicat

Internship Experience

Rubixe, Bengaluru, Karnataka March 2020 – May 2020, Data Science Intern

- **AI for Hiring (Role: AI Engineer)**: Designed business proposal for students' academics and placement data, built model using classification algorithms to predict students' placement, published a research paper in the International Journal of Science and Research (IJSR)

Goa Electronics Limited (GEL), Panaji, Goa - Oct 2018 - Nov 2018, Project Intern

- Worked on the analysis of making compliant government websites, designed Software Requirement Specification (SRS) for labor and employment, land dispute, fire, and emergency services, etc., designed flowcharts for employment exchange services

C-DAC ACTS, Pune, Maharashtra - July 2016 – May 2017, Project Intern

- Worked on Parallelization of Backpropagation algorithm using OpenMP and C in Ubuntu, carried out dissertation work, and presented a paper at the International Workshop on Internet of Things and TV White Spaces (WIOT' 2017)
- Published research paper in the International Journal of Current Research (IJCR)
- Good Knowledge on OpenMP, MPI, Machine Learning, C, Linux Commands

CSIR- National Institute of Oceanography (NIO), Dona Paula, Goa December 2013 – January 2014 as Project Intern

- Worked on Robotic OS and serial programming, and studied different robotic OS like RROS, EmbOS, RTX, RISC, VxWorks, MOOS, etc.
- Implemented a serial port C program to transfer data from one USB port to another in Ubuntu

Education

- Master of Technology – High Performance Computing (Computer Science and Engineering), C-DAC, Pune and Veltech University, Chennai, Tamil Nadu – 2015 – 2017, 8.61 CGPA
- Bachelor of Engineering – Computer Engineering, Goa College of Engineering, Goa University, Goa – 2011 - 2015, 70%
- Higher Secondary School (Class XII) – Science, Goa Board, 2010-2011, 74%
- High School (Class X) – General, Goa Board, 2008-2009, 82.17%

Technical Certifications/Achievements

- Neo4j Certification: Neo4j Fundamentals, Cypher Fundamentals, Graph Data Modeling Fundamentals, Using Neo4j with Python, Intermediate Cypher queries
- Anthropic Claude Certification: Claude 101, Introduction to Agent Skills, Claude Code 101, Claude Code in Action, Introduction to Claude Cowork, Introduction to Model Context Protocol, AI Fluency Framework & Foundations, AI Capabilities and Limitations, AI Fluency for educators, AI Fluency for students, Introduction to subagents, Teaching AI Fluency, AI Fluency for nonprofits, AI Fluency for Small Businesses, Model Context Protocol_ Advanced Topics, Building with the Claude API, Claude Platform 101, Claude with Amazon Bedrock, Claude with Google Cloud's Vertex AI
- Persistent Systems: High Five Individual Awards – Completed assignment on IBM-CMOD

- Persistent Systems: SASVA Immersive Workshop
- Databricks: Knowledge Badge - Get Started with Data Governance on Databricks, Building RAG Agents with Agent Bricks, Deploying and Monitoring Agent Applications on Databricks, Building Reliable Conversational Agents with Genie, Data Governance at Scale
- Databricks: Generative AI Application Evaluation and Governance, Get Started with Data Governance on Databricks and Get Started with Databricks for Data Engineering
- Databricks Delivery Specialization: CDW Migration Best practices, EDW Migration Best practices
- Databricks: Generative AI Application Evaluation and Governance, Generative AI Solution and Application Development, Generative AI Application Deployment and Monitoring
- Persistent Systems: CursorAI Course, GitHub Copilot Course
- Databricks: Knowledge Badge - Generative AI Application Development, EDW-ETL Migration to the Data Intelligence Platform
- Partner Training - Financial Services Industry Specialization for Gen AI & LLM Databricks Badge
- Partner Training - Healthcare & Life Sciences Industry Specialization for Gen AI & LLM Databricks Badge
- Snowflake: Snowflake Sales Professional Accreditation, Snowflake Beginner Fundamentals
- Persistent Systems: iAura Course, Generative AI Foundation Course
- LinkedIn: Preparing for the Future of Work with AI Agents, Preparing Your Organization for AI Agents, Leading through AI Agent Disruption, Transforming Business with AI Agents: Autonomous Efficiency and Decision-Making, Fundamentals of Agentic AI: Business Implications and Ethical Insights, Cybersecurity Foundations
- Persistent Systems: Participated in Semicolons 2026, Annual Global Hackathon of Persistent Systems
- Persistent Systems: Gen AI Competency Level - 3 Certification
- Persistent Systems: Participated in Semicolons 2025, Annual Global Hackathon of Persistent Systems
- Persistent Systems: Python ML Competency Level - 3 Certification, Python ML Competency Level - 2 Certification
- Persistent Systems: Generative AI for Practitioners Certification (Objective and Subjective), GenAI Foundation – GenAI Assisted Coding using Amazon Code Whisperer
- Databricks Accredited Generative AI Fundamentals
- Career Essentials in Generative AI by Microsoft and LinkedIn
- Databricks Lakehouse Machine Learning Associate
- Persistent Systems: Participated in Semicolons 2023, Annual Global Hackathon of Persistent Systems
- Fundamentals of the Databricks Lakehouse Platform Accreditation
- Persistent Systems: Information Security Awareness Secure Coding/Data Handling & Privacy Training
- IABAC, Amsterdam: Certified Data Scientist & Data Science Foundation by
- Deep Learning specialization by Coursera and deeplearning.ai on Coursera.org
- Certified Data Scientist course completion certificate by DataMites, Bengaluru
- Paper publication “Parallelization of Backpropagation Algorithm & Benchmarking” in IJCR
- Paper publication “Artificial Intelligence for Hiring” in IJSR
- Paper presentation on “Backpropagation Algorithm & Use of OpenMP in ML” in International Workshop on IOT & TV White Spaces
- Udemy: Agentic AI Playbook Complete Guide for Tech Leaders
- Udemy: Context Engineering Masterclass LLMs, RAG & Agents
- Udemy: Agentic Workflows Design End-to-End AI Processes
- Udemy: Master Class for Agentic AI Design Patterns
- Udemy: LangChain & LLMs - Build Autonomous AI Tools Masterclass
- Udemy: Prompt Engineering for Everyone Bootcamp
- Udemy: Generative AI with Context: RAG, CAG & KAG Applications
- Udemy: Agentic AI Production Grade AI Agents using CrewAI and AWS
- Udemy: Codex - The Practical Guide
- Udemy: Introduction to Generative AI and LLMs
- Udemy: Foundations in Responsible AI & AI Ethics
- Udemy: QC101 Quantum Computing & Intro to Quantum Machine Learning
- Udemy: Building AI Agents & Agentic AI System via Microsoft Autogen
- Udemy: Ultimate RAG Bootcamp Using LangChain, LangGraph & LangSmith
- Udemy: Building Gen AI App 12+ Hands-on Projects with Gemini Pro
- Udemy: Complete MCP Bootcamp- Build Next-Gen AI Agents with MCP
- Udemy: Full stack generative and Agentic AI with python
- Udemy: MCP Masterclass - Complete Guide to MCP in Python [2025]
- Udemy: Intro to AI Agents and Agentic AI

- Udemy: LangGraph- Develop LLM powered AI agents with LangGraph
- Udemy: Agentic AI for Beginner
- Udemy: Terraform on Azure
- Udemy: Terraform on AWS for Beginners Real-World Project
- Udemy: Terraform for the Absolute Beginners with Labs
- Udemy: 2026 Bootcamp Generative AI, LLM Apps, AI Agents, Cursor AI
- Udemy: The Complete Prompt Engineering for AI Bootcamp (2025)
- Udemy: Intro to MCP
- Udemy: 2025 - Understand, Test, Fine-tune AI Model with HuggingFace
- Udemy: Generative AI solutions with Azure OpenAI
- Udemy: Learn Hugging Face Bootcamp
- Udemy: Mastering Generative AI with PyTorch - Hands-on Experience
- Udemy: Evaluating AI Agents
- Udemy: Chatbot - The Development Guide 2025 (Beginner + Advanced)
- Udemy: Basic to Advanced - Retrieval-Augmented Generation (RAG)
- Udemy: Langchain for beginners - Build GenAI LLM Apps in Easy Steps
- Udemy: LangGraph in Action - Develop Advanced AI Agents with LLMs
- Udemy: Intro to Google's A2A Protocol - Interoperable AI Agents
- Udemy: AI Agentic Design Patterns with Ollama & OpenAI Guide
- Udemy: Amazon Q Developer for Programmers and DevOps AWS AI coding
- Udemy: Responsible AI - Principles, Practices, and Applications
- Udemy: Prompt Engineering for Beginners - Learn ChatGPT Prompting
- Udemy: LLM Fine Tuning Fundamentals + Fine tune OpenAI GPT model
- Udemy: LLMs Mastery - Complete Guide to Transformers & Generative AI
- Udemy: Generative AI for Beginners
- Udemy: Intermediate Python - Memory, Decorator, Async, Cython & more
- Udemy: Learn Selenium with Python, PyTest & Frameworks
- Udemy: Serverless using AWS Lambda for Python Developers
- Udemy: AWS MasterClass - Monitoring and DevOps with AWS CloudWatch
- Udemy: AWS Security Best Practices
- Udemy: Introduction to Virtualization - 90 Minute Crash Course
- Udemy: AWS EC2 Crash Course + Load Balancing with Demos and PDFs
- Udemy: Learn Basic DB2 on Mainframe for Beginners
- Udemy: DB2 For Beginners
- Udemy: Containers on AWS ECS, EKS, and Fargate 2 hour Crash Course
- Udemy: AWS Serverless Lambda Functions 2 hr. Hands On Crash Course
- Udemy: Hands-On Introduction to CloudFormation - 2 hour course
- Udemy: Amazon Aurora PostgreSQL from A to Z
- Udemy: Git Going Fast One Hour Git Crash Course
- Udemy: Complete Agentic AI Bootcamp with LangGraph and Langchain
- Udemy: AWS DynamoDB 2025: High Performance Strategies Included
- Udemy: Complete Generative AI Course With Langchain and Huggingface
- Udemy: AWS RDS & Aurora Crash Course – Zero to Hero in 3 Hours!
- Udemy: Master RAG: Ultimate Retrieval-Augmented Generation Course
- Udemy: AWS Certified Machine Learning Engineer Associate: Hands On
- Udemy: The Absolute Beginners Guide to Cyber Security 2025 - Part 1, 2, 3 and 4
- Udemy: Introduction to Cyber Security - 2 Hour Crash Course
- Udemy: AWS CDK for professionals (Python and TypeScript)
- Udemy: AWS SAM Framework and AWS Lambda - A Complete Hands-on Guide
- Udemy: AWS Lambda - A Practical Guide - Learn from an Expert
- Udemy: Learn AWS Identity Management with AWS IAM, SSO & Federation
- Udemy: AWS Identity Access Management (IAM) Practical Applications
- Udemy: Introduction to AWS - Understand AWS basics in 3 hours
- Udemy: Gemini Google AI: The All-in-One AI Masterclass 2025
- Udemy: Mathematics Behind Large Language Models and Transformers
- Udemy: Introduction to Large Language Models (LLMs) In Python
- Udemy: MCP Crash Course Complete Model Context Protocol in a Day
- Udemy: Intro to Large Language Models (LLMs)

- Udemy: Automate the Boring Stuff with Python Programming
- Udemy: Complete Python Bootcamp 2024: Zero to Expert in Python
- Udemy: AWS SageMaker Practical for Beginners Build 6 Projects
- Udemy: MongoDB – The Complete Developer's Guide 2024
- Udemy: Complete Tensorflow2 and Keras Deep Learning Bootcamp
- Udemy: Taming Big Data with Apache Spark and Python – Hands-on
- Udemy: Natural Language Processing with Python
- Udemy: Natural Language Processing with Transformers in Python
- Udemy: LangChain with Python Bootcamp
- Udemy: REST APIs with Flask and Python in 2024
- Udemy: Python for Data Science and Machine Learning Bootcamp
- Udemy: Complete Data Science, Machine Learning, DL, NLP Bootcamp
- Udemy: Git for Geeks: Quick Git Training for Developers
- Udemy: Amazon Bedrock & AWS Generative AI [Beginner to Advanced]
- Udemy: Generative AI: Beginner to Pro with OpenAI & Azure OpenAI
- Udemy: LangChain- Develop LLM-powered applications with LangChain
- Udemy: Deployment of Machine Learning Models
- 1st prize in National Summer Training Program on "Bigdata & Hadoop" by Revert Technology, EDC-IIT Roorkee
- Diploma Course in Java programming with JDBC and C programming at MICE

Non-Technical Certifications

- Udemy: Artificial Intelligence and Business Strategy AI Roadmap
- Udemy: Become a 10x Product Manager with ChatGPT & Generative AI
- Udemy: The EU AI Act Explained: Compliance and Practical Impact
- Udemy: Advanced Product Management – Vision, Strategy and Metrics
- Udemy: AI Ethics/Responsible Use
- Udemy: AI Leader: Generative AI & Agentic AI for Leaders & Founders
- Udemy: Mastering Workplace Communication Skills
- Udemy: AI for Product Management & Innovation
- Udemy: Product Management & Product Design with Generative AI
- Udemy: Generative AI for Leaders: Strategy & Innovation
- Udemy: Project Management with Gen AI: A Guide for Project Managers
- Udemy: Business Communication Skills: Business Writing & Grammar
- Udemy: Generative AI for Leaders & Business Professionals
- Udemy: Entrepreneurship - How To Start A Business From An Idea
- Udemy: Generative AI for Business Leaders: A Quick Overview
- Udemy: Product Management 101
- Udemy: Project Management Fundamentals
- Udemy: Leadership: Practical Leadership Skills
- Udemy: Business Communication Skills: Handling Clients & Customers
- Udemy: Communication Skills for Professionals
- Udemy: Communication Skills: Become Clear, Concise, Confident
- Udemy: Delegation Strategies for Managers
- AMCAT certification exam - 77.72 percentile, Apr 2017

Declaration

I hereby declare that all the information provided above is true to my knowledge.

Sign: **Date**: